

Numbers Three Ways

Answer Key

Grade 2 Math

Quick Answers

- | | |
|-------------------|--------------------------|
| 1. $40 + 6$ | 6. $700 + 5$ |
| 2. 329 | 7. — |
| 3. $500 + 70 + 3$ | 8. 839 |
| 4. — | 9. 907 |
| 5. 641 | 10. missing part = 60, — |
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What is verified

7 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 4, 7, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 2 Math

2.NBT.A – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–4 of 5 across 11 tagged checks.

Common wrong answers

- 6. *If they got 750: moved the 5 into the tens place because of the zero*
 - 8. *If they got 983: wrote the digits in the order given instead of by place value*
 - 9. *If they got 97: dropped the zero placeholder and wrote a two-digit number*
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1. Write 46 in expanded form.

Solution: 46 has 4 tens and 6 ones. The 4 tens are worth 40 and the 6 ones are worth 6, so $46 =$ $40 + 6$

2. Write the number in standard form: $300 + 20 + 9$

Solution: Put each part in its place: 3 hundreds, 2 tens, 9 ones. Reading the places together gives 329

3. Write 573 in expanded form.

Solution: 573 has 5 hundreds (500), 7 tens (70), and 3 ones (3), so $573 =$

$500 + 70 + 3$

4. Write 285 in word form.

Solution: Say the hundreds, then the rest: 2 hundreds is “two hundred”; 85 is “eighty-five”. Model answer: *two hundred eighty-five* (hyphen optional; no “and”).

5. Write the number in standard form: six hundred forty-one

Solution: “six hundred” is 6 hundreds, “forty” is 4 tens, “one” is 1 one: $600 + 40 + 1$.

641

6. Write 705 in expanded form.

Solution: 705 has 7 hundreds, *zero* tens, and 5 ones. The empty tens place contributes nothing, so only two parts are written: $705 =$

$700 + 5$

7. Write 460 in word form.

Solution: 4 hundreds is “four hundred”; 60 is “sixty”; there are no ones to say. Model answer: *four hundred sixty*.

8. Write the number in standard form: $9 + 800 + 30$

Solution: Sort the parts by place first: 800 is the hundreds part, 30 is the tens part, 9 is the ones part. That is 8 hundreds, 3 tens, 9 ones — not nine hundred anything!

839

9. Write the number in standard form: nine hundred seven

Solution: “nine hundred” is 9 hundreds and “seven” is 7 ones. No tens are named, so the tens place gets a zero placeholder:

907

10. Challenge: $800 + \underline{\quad} + 2 = 862$, and explain.

Solution: The number 862 has 8 hundreds (800, already shown), 6 tens, and 2 ones (2, already shown). The missing part is the tens: 6 tens =

60

What a good explanation contains: the 800 and the 2 already cover the hundreds and ones places, so the missing part must come from the tens place; the tens digit of 862 is 6, which is worth 60.